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Beyond Ridge Regression: Enhancing Distribution of Relaxation Times Deconvolution

Baptiste Py[1] and Francesco Ciucci[2,3,z]

[1]*Department of Mechanical and Aerospace Engineering, The Hong Kong University of Science and Technology, Hong Kong SAR, People's Republic of China*
[2]*Chair of Electrode Design for Electrochemical Energy Systems, University of Bayreuth, 95448 Bayreuth, Germany*
[3]*University of Bayreuth, Bavarian Center for Battery Technology (BayBatt), Universitätsstraße 30, 95447 Bayreuth, Germany*

The distribution of relaxation times (DRT) has emerged as a promising method for analyzing electrochemical impedance spectroscopy (EIS) data. The standard approach for reconstructing the DRT from measured impedances consists of regularized regression, which usually leverages the Euclidean norm. In this work, we show for the first time that the 1-norm is often more accurate than ridge regression and the infinity-norm. We also demonstrate that the 1-norm is more robust against discontinuities in the DRT and outliers in the impedance data. Overall, this work is expected to enhance regularized regression of non-parametric methods when analyzing EIS spectra.
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Electrochemical impedance spectroscopy (EIS) is a powerful and versatile technique widely used in the study of electrochemical devices and systems.[1] To overcome the limitations of equivalent circuits and physical models when analyzing EIS data,[2,3] non-parametric methods, such as the distribution of capacitive times,[4] distribution of diffusion times,[5] and distribution of relaxation times (DRT),[6] have been developed. Despite the more and more widespread usage of the DRT method,[7–9] extracting meaningful distributions from measured EIS spectra remains challenging due to the ill-posed nature of the inverse problem, leading to non-unique solutions and strong sensitivity to experimental errors.[10,11] Among the methods used to recover the unknown DRT,[12–14] regularized regression is the most widespread approach, which formulates DRT inversion as the solution of[15,16]

$$\hat{\boldsymbol{x}} = \underset{\boldsymbol{x}}{\operatorname{argmin}}\,[d(\boldsymbol{Z}_{\exp}, \boldsymbol{Z}_{\mathrm{DRT}}(\boldsymbol{x}))^{\alpha} + \lambda \cdot P(\boldsymbol{x})] \qquad [1]$$

where $\boldsymbol{x} = (L_0, R_\infty, \boldsymbol{\gamma})^{\top}$ is a vector containing the inductance (L_0), ohmic resistance (R_∞), and the DRT discretization ($\boldsymbol{\gamma}$), $d(\bullet, \bullet)$ is a distance, $\boldsymbol{Z}_{\exp}$ is the vector of the experimental impedances, $\boldsymbol{Z}_{\mathrm{DRT}}(\boldsymbol{x})$ is the vector of the model (DRT-based) impedance, $P(\boldsymbol{x})$ is a penalty function, and α and λ are two positive scalars.[17]

Ridge regularization (RR) typically employs the Euclidean distance between the measured and model impedance, i.e., $d(\boldsymbol{Z}_{\exp}, \boldsymbol{Z}_{\mathrm{DRT}}(\boldsymbol{x})) = \|\boldsymbol{Z}_{\mathrm{DRT}}(\boldsymbol{x}) - \boldsymbol{Z}_{\exp}\|_2$, with $\alpha = 2$, and sets the penalty $P(\boldsymbol{x}) = \|\boldsymbol{L}\boldsymbol{x}\|_2^2$ ($\boldsymbol{L}$ is either the identity or a differential matrix).[5] This notation suggests that other distances/norms could be used. For example, it is well known that using the 1-distance in linear regression enhances robustness against outliers in $\boldsymbol{Z}_{\exp}$.[10] Furthermore, we could select $P(\boldsymbol{x}) = \|\boldsymbol{L}\boldsymbol{x}\|_q$. Lasso regression, for instance, utilizes the Euclidean distance with $\alpha = 2$ and $P(\boldsymbol{x}) = \|\boldsymbol{x}\|_1$.[18]

This article extends existing regularized regression methods for EIS data analysis by investigating alternative distance metrics and regularization norms. Specifically, we examine the four combinations arising from 1-norm and ∞-norm. Empirical analysis reveals that the 1–1 pair (1-norm for both experimental-model distance and the $\|\boldsymbol{L}\boldsymbol{x}\|_1$ regularization term) yields superior DRT recovery compared to conventional RR. Additionally, the 1–1 pair demonstrates increased robustness to outliers and excels when recovering discontinuous DRTs, such as the Gerischer[19] and piecewise-constant[20] models. This research offers a new approach for enhancing the accuracy of DRT recovery in EIS analysis.

Theoretical and Experimental Methods

Choosing the p and q norm pairs with $\alpha = 1$ allows us to reformulate[1] and obtain the corresponding estimate, $\hat{\boldsymbol{x}}_{p,q}$, as[a]

$$\hat{\boldsymbol{x}}_{p,q} = \underset{\boldsymbol{x}}{\operatorname{argmin}}\,(\|\boldsymbol{A}\boldsymbol{x} - \boldsymbol{Z}_{\exp}\|_p + \lambda\,\|\boldsymbol{L}\boldsymbol{x}\|_q) \qquad [2]$$

where $\boldsymbol{A}$ is the matrix discretizing the DRT problem.[4]

To quantify the deviation of $\hat{\gamma}(\log\tau)$ from the exact $\gamma(\log\tau)$, we utilized the following two errors:[b]

$$\varepsilon_{\gamma} = \|\boldsymbol{\gamma}(\log\boldsymbol{\tau}) - \hat{\boldsymbol{\gamma}}(\log\boldsymbol{\tau})\|_2 \approx \|\gamma(\log\tau) - \hat{\gamma}(\log\tau)\|_2 \qquad [3a]$$

$$\varepsilon_{\partial\gamma} = \left\| \frac{d\boldsymbol{\gamma}}{d\log\tau}(\log\boldsymbol{\tau}) - \boldsymbol{L}_1\hat{\boldsymbol{x}} \right\|_2 \approx \left\| \frac{d\gamma}{d\log\tau} - \frac{d\hat{\gamma}}{d\log\tau} \right\|_2 \qquad [3b]$$

where $\boldsymbol{\gamma}(\log\boldsymbol{\tau}) = (\gamma(\log\tau_1), \gamma(\log\tau_2), \ldots, \gamma(\log\tau_N))^{\top}$ and $\hat{\boldsymbol{\gamma}}(\log\boldsymbol{\tau}) = (\hat{\gamma}(\log\tau_1), \hat{\gamma}(\log\tau_2), \ldots, \hat{\gamma}(\log\tau_N))^{\top}$ are the exact and regressed DRTs, respectively, discretized at the colocation points $\log\tau_1$, $\log\tau_2$, …, $\log\tau_N$, and $\boldsymbol{L}_1$ is the first-order differentiation matrix. The impedance error, ε_Z, was defined as $\varepsilon_Z = \|\boldsymbol{Z} - \boldsymbol{A}\hat{\boldsymbol{x}}_{p,q}\|_2$ with $\boldsymbol{Z}$ being the vector of the exact impedance, and the average error as $\overline{\varepsilon} = \frac{1}{N_{\exp}}\sum_{n=1}^{N_{\exp}} \varepsilon_n$ with $N_{\exp} = 100$. We note that ε_{γ} and $\varepsilon_{\partial\gamma}$ are used to estimate the similarity between the recovered DRT and its first derivative to their exact counterparts.

To assess the quality of the recovered DRTs, we employed synthetic experiments. We utilized well-defined models, such as the 1 × ZARC, 2 × ZARCs (separated, distant, and close), Gerischer, piecewise-constant, Havriliak-Negami, and fractal models (see Table S1 of the Supplementary Information (SI) for the values of the circuit parameters). Frequencies were spaced logarithmically between 10 mHz and 1.0 MHz with 10 points per decade. These models were then corrupted with Gaussian noise ($\sigma_n^{\exp} = 0.2\ \Omega$).[4] 100 synthetic experiments were generated for each model. Discretization of the DRT was achieved using piecewise linear

[a]In our case, we use $p, q = 1$ or ∞, which means solving [2] requires simple linear programming.
[b]We note that the definition of $\varepsilon_{\partial\gamma}$ in [3b] is improper for discontinuous DRTs (Comparing regression distances section).

[z]E-mail: francesco.ciucci@uni-bayreuth.de

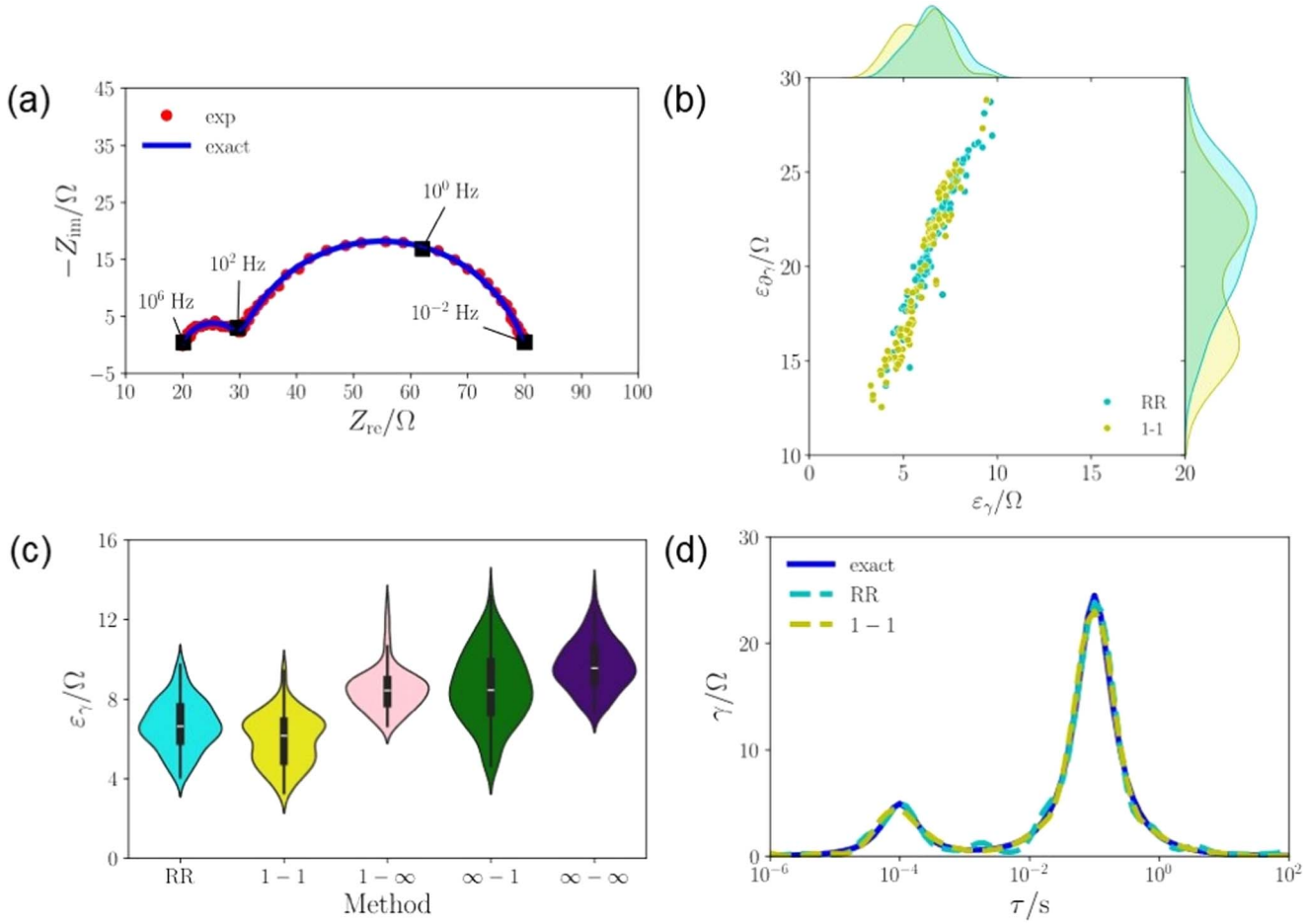

Figure 1. Distant 2 × ZARC model; (a) Nyquist plots of the exact and experimental impedances; (b) joint plots of the distributions of ε_γ and $\varepsilon_{\partial\gamma}$ for RR and the 1–1-pair; (c) violin plots of ε_γ for RR and each pair; (d) optimal DRTs recovered with RR and the 1–1 pair with the exact DRT also shown.

Table I. For each impedance model, average values of ε_γ and $\varepsilon_{\partial\gamma}$. The lowest values in each category are bolded[a)] and the lowest values for ε_γ are shown in red.

Impedance model	1 × ZARC	Separated 2 × ZARC	Distant 2 × ZARC	Close 2 × ZARC	Gerischer	Piecewise-constant	Havriliak-Negami	Fractal
				Average error $\bar{\varepsilon}_\gamma/\Omega$				
RR	5.89×10^0	$\mathbf{7.83 \times 10^0}$	6.67×10^0	7.29×10^0	2.80×10^1	$\mathbf{8.64 \times 10^0}$	2.63×10^0	3.46×10^1
$1-1$ (L_3)	$\mathbf{5.22 \times 10^0}$	7.97×10^0	$\mathbf{5.94 \times 10^0}$	$\mathbf{7.27 \times 10^0}$	$\mathbf{2.72 \times 10^1}$	9.78×10^0	$\mathbf{2.10 \times 10^0}$	$\mathbf{3.38 \times 10^1}$
$1-\infty$ (L_3)	7.12×10^0	1.02×10^1	8.48×10^0	9.12×10^0	2.90×10^1	1.05×10^1	3.76×10^0	3.45×10^1
$\infty-1$ (L_3)	7.29×10^0	1.13×10^1	8.58×10^0	1.16×10^1	2.77×10^1	1.41×10^1	3.44×10^0	3.52×10^1
$\infty-\infty$ (L_3)	8.33×10^0	1.23×10^1	9.69×10^0	1.20×10^1	3.01×10^1	1.44×10^1	4.43×10^0	3.57×10^1
				Average error $\bar{\varepsilon}_{\partial\gamma}/\Omega$				
RR	1.97×10^1	$\mathbf{2.64 \times 10^1}$	2.15×10^1	$\mathbf{2.44 \times 10^1}$	1.39×10^2	$\mathbf{4.64 \times 10^1}$	5.55×10^0	$\mathbf{1.58 \times 10^2}$
$1-1$ (L_3)	$\mathbf{1.79 \times 10^1}$	2.67×10^1	$\mathbf{1.96 \times 10^1}$	2.47×10^1	$\mathbf{1.38 \times 10^2}$	5.19×10^1	$\mathbf{4.43 \times 10^0}$	1.58×10^2
$1-\infty$ (L_3)	2.40×10^1	3.42×10^1	2.76×10^1	3.07×10^1	1.42×10^2	5.64×10^1	9.05×10^0	1.59×10^2
$\infty-1$ (L_3)	2.29×10^1	3.47×10^1	2.54×10^1	3.60×10^1	1.38×10^2	6.56×10^1	6.49×10^0	1.58×10^2
$\infty-\infty$ (L_3)	2.58×10^1	3.78×10^1	2.92×10^1	3.67×10^1	1.42×10^2	6.76×10^1	9.32×10^0	1.59×10^2
				Average error $\bar{\varepsilon}_\gamma/\Omega$				
$1-1$ (L_0)	5.54×10^1	7.88×10^1	5.66×10^1	8.76×10^1	4.87×10^1	6.65×10^1	4.21×10^1	4.88×10^1
$1-1$ (L_1)	1.40×10^1	1.98×10^1	1.45×10^1	1.67×10^1	3.35×10^1	$\mathbf{8.04 \times 10^{-1}}$	7.79×10^0	4.11×10^1
$1-1$ (L_2)	$\mathbf{5.16 \times 10^0}$	$\mathbf{7.77 \times 10^0}$	$\mathbf{5.74 \times 10^0}$	7.87×10^0	$\mathbf{2.66 \times 10^1}$	8.26×10^0	2.39×10^0	$\mathbf{3.30 \times 10^1}$
$1-1$ (L_3)	5.22×10^0	7.97×10^0	5.94×10^0	$\mathbf{7.27 \times 10^0}$	2.72×10^1	9.78×10^0	$\mathbf{2.10 \times 10^0}$	3.38×10^1
				Average error $\bar{\varepsilon}_{\partial\gamma}/\Omega$				
$1-1$ (L_0)	5.20×10^1	7.89×10^1	5.30×10^1	6.65×10^1	1.53×10^2	1.16×10^2	2.86×10^1	1.71×10^2
$1-1$ (L_1)	7.02×10^1	1.02×10^2	7.19×10^1	8.51×10^1	1.73×10^2	$\mathbf{4.77 \times 10^0}$	3.51×10^1	1.76×10^2
$1-1$ (L_2)	1.96×10^1	2.86×10^1	2.07×10^1	3.02×10^1	$\mathbf{1.25 \times 10^2}$	4.40×10^1	7.15×10^0	$\mathbf{1.55 \times 10^2}$
$1-1$ (L_3)	$\mathbf{1.79 \times 10^1}$	$\mathbf{2.67 \times 10^1}$	$\mathbf{1.96 \times 10^1}$	$\mathbf{2.47 \times 10^1}$	1.38×10^2	5.19×10^1	$\mathbf{4.43 \times 10^0}$	1.58×10^2

a) We note that the average of $\varepsilon_{\partial\gamma}$ values for discontinuous DRTs (e.g., Gerischer, piecewise-constant, and fractal models) must be interpreted with caution. These DRTs are inherently non-differentiable due to their discontinuities, rendering the calculation of $\varepsilon_{\partial\gamma}$ inappropriate within a strict mathematical context.

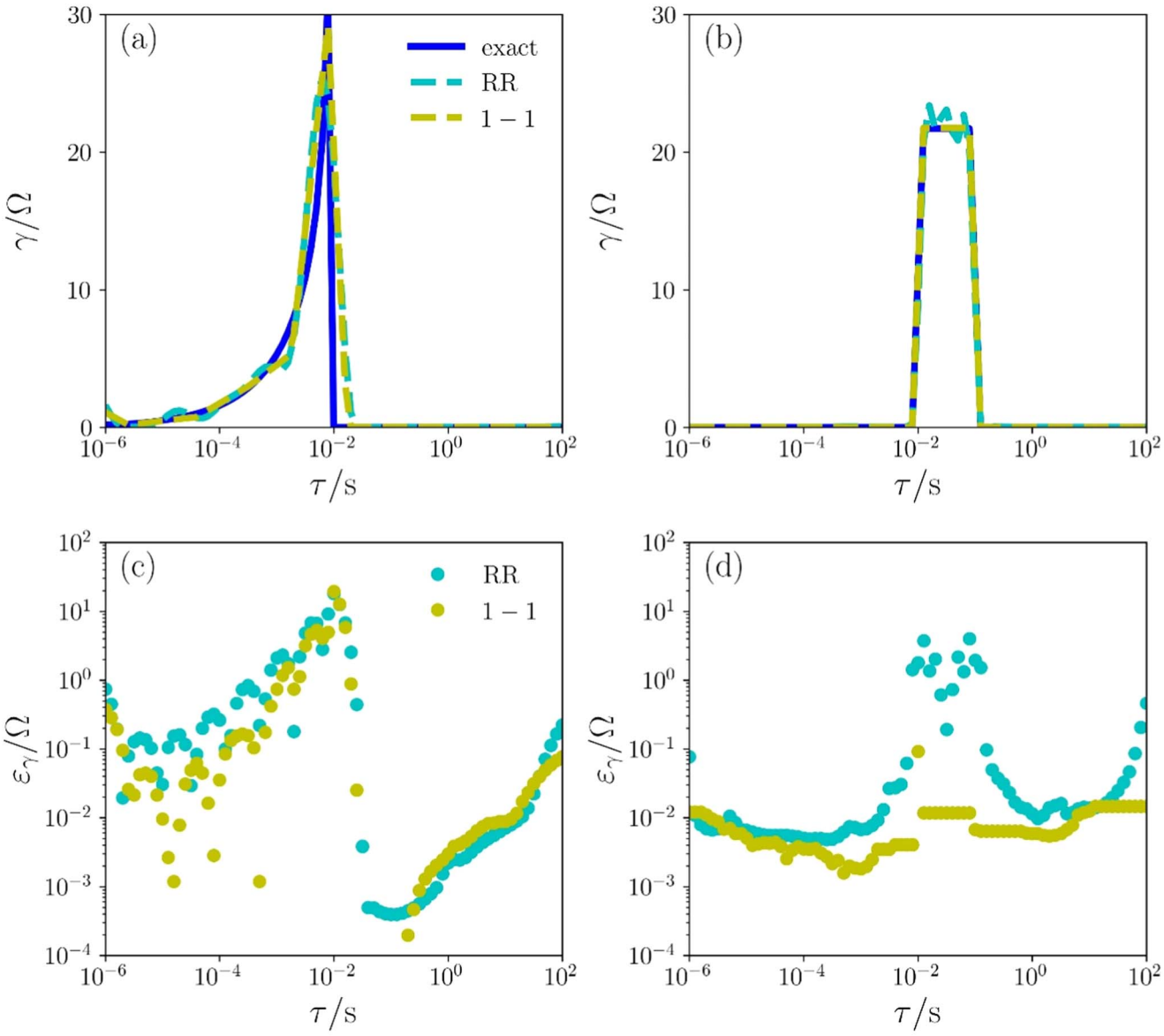

Figure 2. The optimal DRTs recovered with RR and the 1–1 pair are shown with the exact DRT for the (a) Gerischer ($\boldsymbol{L}_2$ matrix) and (b) piecewise-constant ($\boldsymbol{L}_1$ matrix) models. The corresponding errors ε computed with the average DRTs recovered with RR and the 1–1 pair are shown in panels (c) and (d).

functions,[4] and the timescales were chosen to be the inverse of the frequencies. The $\hat{\boldsymbol{x}}_{p,q}$ of each synthetic experiment was obtained using CVXPY with the GUROBI optimizer.[21] To find the best regularization parameter λ, we employed Bayesian optimization and minimized ε_γ (Fig. S1a). Unless otherwise specified, a third-order differentiation matrix, $\boldsymbol{L}_3$, was used to solve [2]. To assess robustness against outliers (Robustness against noise and outliers section), we used the distant $2 \times$ ZARC model and arbitrarily set the experimental noise at the frequencies 0.01, 1.0, and 10^4 Hz to be 25 times higher than the baseline white noise (Fig. 1a). Lastly, the impedances were estimated as $\boldsymbol{A}\hat{\boldsymbol{x}}_{p,q}$.

Results and Discussion

Synthetic experiments.— *Comparing regression distances.*—As shown in Table I (the best values in each category are shown in bold font, and the best overall values are in red) and Fig. S1, the 1–1 pair consistently achieved the lowest error values, as defined in Ref. 3. Notably, the error ε_γ for the distant $2 \times$ ZARC model (Fig. 1a) was significantly lower with the 1–1 pair compared to RR and other pairs. This finding aligns with the joint plot of ε_γ and $\varepsilon_{\partial\gamma}$ (Fig. 1b) and the ε_γ violin plot (Fig. 1c). Figure 1d shows the DRTs recovered using the 1–1 pair and RR (with optimal λ); the 1–1 pair eliminated oscillations evident in the RR-recovered DRT. Furthermore, the average impedance recovered with the 1–1 pair closely approximated the exact impedance for the distant $2 \times$ ZARC model (Fig. S2 and average impedance error $\overline{\varepsilon}_Z = 9.04 \times 10^{-1}\ \Omega$). These preliminary results strongly suggest that the 1–1 pair outperforms RR and all other considered pairs for an optimal $\boldsymbol{L}$-matrix choice (Comparing differentiation matrices in the penalty term section). Subsequent analysis will, therefore, focus primarily on the 1–1 pair, which will be compared to RR.

Comparing differentiation matrices in the penalty term.—We now discuss the influence of the differentiation matrix, $\boldsymbol{L}$, on the performance of the 1–1 pair. We employed 0th-order ($\boldsymbol{L}$ is the identity matrix), first-, second-, and third-order differentiation matrices to recover the DRT from Ref. 2 ($p = q = 1$). For all

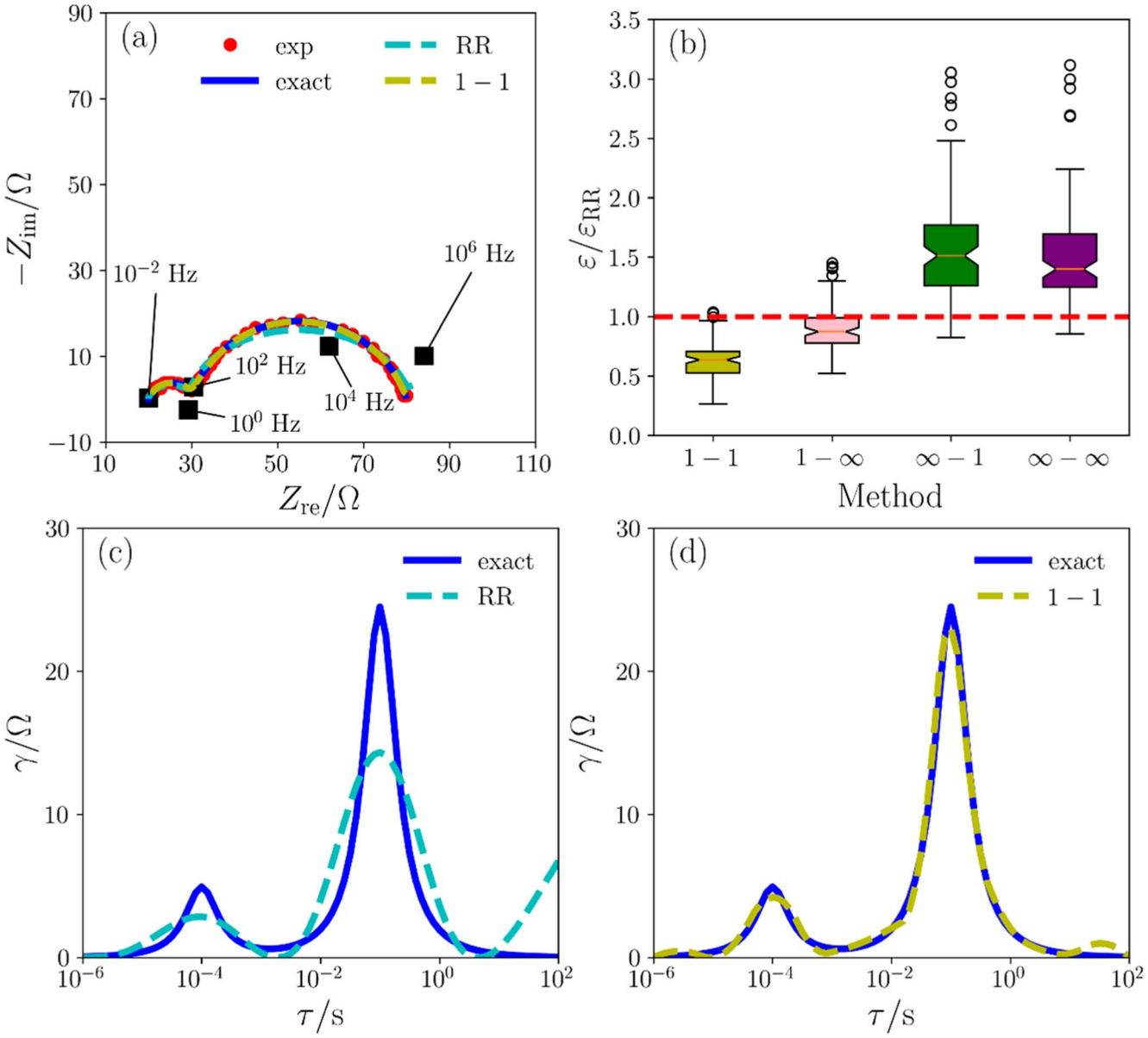

Figure 3. Distant 2 × ZARC model and three outliers; for one synthetic experiment, (a) Nyquist plots of the impedances recovered with RR and the 1–1 pair with the exact and experimental impedances also shown; (b) boxplots of the total DRT error, ε, divided by the error for RR, $\varepsilon_{\mathrm{RR}}$; exact DRT and average DRT recovered with (c) RR and (d) the 1–1 pair.

DRTs except the piecewise-constant DRT, the 1–1 pair achieved the best accuracy with $\boldsymbol{L}_2$ or $\boldsymbol{L}_3$. These higher-order differentiation matrices promote regularity in the recovered DRT. Using $\boldsymbol{L}_2$ promotes a piecewise linear $\gamma(\log\tau)$ analogous to lasso-type regularization for the second derivative (Fig. 2a). Conversely, $\boldsymbol{L}_3$ acts as a lasso-type regularizer for the third derivative, resulting in smoother solutions with no jumps in the second derivative (Fig. 1). Regarding the piecewise-constant DRT, the $\boldsymbol{L}_1$ matrix led to the closest recovery of the exact DRT because this matrix favors piecewise constant solutions (Fig. 2b). Compared to RR, the 1–1 model results in better adherence to the exact DRT (Figs. 2a and 2b), as evidenced by the smaller errors of Figs. 2c and 2d.

Robustness against noise and outliers.—As discussed in introduction section, regression using the 1 distance is generally more robust against outliers than the Euclidean norm.[22] To illustrate this in the context of EIS, we used the distant 2 × ZARC impedance to which we added white noise and three outliers (Theoretical and Experimental Methods section and Fig. 3a).

The 1–1 pair ($\bar{\varepsilon} = 2.59 \times 10^{1}\ \Omega$) demonstrated better outlier management than the RR approach ($\bar{\varepsilon} = 4.35 \times 10^{1}\ \Omega$), as shown in Fig. 3b. This result is consistent with the understanding that the 1-norm is generally more suitable for robust regression than the 2-norm.[23] The DRTs obtained with RR approximately identified each peak and often diverged at high timescales (Fig. 3c). Conversely, the DRTs recovered with the 1–1 pair were much closer to the exact DRT (Fig. 3d). The exact impedance was also more accurately recovered with the 1–1 pair than RR (Figs. 3a and S2, and $\bar{\varepsilon}_Z = 3.68 \times 10^{0}\ \Omega$ and $9.84 \times 10^{-1}\ \Omega$ for RR and the 1–1 pair, respectively).

Real experiment.—Finally, we demonstrated the applicability of the 1–1 pair to real EIS data collected from our previous experiments.[24] In parallel to the DRT estimations, an equivalent circuit model composed of two ZARC elements in series with an inductor and resistor was used to regress the experimental impedance. Our results showed that the DRT ($\varepsilon = 1.36 \times 10^{0}\ \Omega$) and

impedance ($\varepsilon_Z = 2.78 \times 10^{-1}\ \Omega$) recovered with the 1–1 pair were more accurate matches to the equivalent-circuit-fitted DRT and impedance compared to the RR-recovered DRT ($\varepsilon = 1.43 \times 10^{0}\ \Omega$) and impedance ($\varepsilon_Z = 2.84 \times 10^{-1}\ \Omega$), see Fig. S2.

Conclusions

Though now widely used for analyzing EIS data, DRT deconvolution remains challenging. Regularized regression is a standard method for obtaining the DRT but can be improved by fine-tuning the selection of suitable distances and norms. Our work demonstrates that directly regressing impedance data against the DRT model, employing the 1-distance for the data/model discrepancy and the 1-norm for the penalty, generally yields higher accuracy than conventional RR and other studied pairs comprising 1 and infinity norms, particularly for discontinuous DRTs and datasets containing outliers. This novel approach offers a promising avenue for further exploration, including the investigation of additional penalties and extensive analysis using real-world data.

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ORCID

Baptiste Py https://orcid.org/0000-0003-1294-6138
Francesco Ciucci https://orcid.org/0000-0003-0614-5537

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